

GitSkills: dataset profiling

Muhammad Shafay Tanveer Hassan - V1

Summary

For Assignment 1, I adapted the task to the GitSkills dataset of the MSR 2027 Mining Challenge and ran it on all 3.8M files. Most skills are never copied, but a few big repos hold a large part of all the files. Skills with scripts are only slightly more likely to be copied into many repos.

- The dataset has 3,797,117 SKILL.md files from 282,200 public GitHub repos (July 2026). Many files are the same, so there are only 1,877,981 different skills. 50.5% of the files are exact copies of another file.
- Most skills are never copied, 79.3% appear only once. But a few repos hold a lot of files. 312 repos (0.11%) hold 36.7% of all skill files.
- 11.4% of skills come with scripts, mostly Python, TypeScript, JavaScript and Shell. 25.9% come with reference files. The bundled files add up to 200.8 GB on GitHub (the dataset itself is about 13 GB), and almost half of them are Markdown.
- Skills with scripts are only slightly more likely to be copied into many repos. Outside the big collection repos, 0.81% of script skills and 0.70% of the other skills are in 10 or more repos (odds ratio 1.15, 95% CI 1.08 to 1.23). The result changes with how I define a collection repo.
- The big collection repos do most of the copying. About three quarters of the copies of widely spread skills are in repos with 100 or more skill files. I checked one security skills collection of 790 skills: 17 other repos hold at least half of it, and 16 of them hold 72% to 99%.

1. Data and method

Data. I used GitSkills (Destefanis, Graziotin, Vaccargiu and Ortu, arXiv:2608.10906). It is a lower bound on how many skills exist. A skill is a folder with a SKILL.md file. The file has instructions for an AI coding agent, and the folder can also have scripts and reference files. There is no registry or package manager, so people share skills by copying folders.

Method. I wrote the code on the authors' 13,000 skill sample first. Then I ran it on the full data from the Parquet files, using DuckDB in Google Colab. My row counts match the paper exactly. For group comparisons I report odds ratios with 95% intervals.

For the text (URLs, language, vocabulary) I used a 1/40 sample of the different skills, picked by content hash. That is 47,160 skills, and I ran language detection on the first 20,000 of them.

These are the terms I use in this report.

Term	What it means
File occurrence	One SKILL.md file in one repo. There are 3,797,117.
Skill	All files with identical content count as one skill. There are 1,877,981.
Copy	A file whose content is identical to another file.
Representative	The one copy of a skill that has the text, front matter and folder details.
Bundled file	A file or folder inside a skill's folder, such as scripts, references or images.
Author	An anonymised commit author code. It can be a bot.
Spread	The number of different repos that hold the same skill.
Collection repo	A repo with 100 or more skill files.

2. Schema

The data has four tables. Only one copy of each skill (the representative) has the text and the folder details.

Table	Rows	One row is	Main columns
artifacts	3,797,117	a SKILL.md file occurrence	repo, path, location class, content hash, representative flag. For the representative of each content: text, front matter, body size, has_scripts, has_references. History columns for a subset.
artifact_siblings	7,264,865	a file or folder bundled with a representative skill	entry name, type, size, text (if small)
repos	282,200	a repository	owner, stars, forks, language, license, dates
mining_runs	7	a collection run	query, timestamps, result count

The 1,877,981 representatives have the text, front matter and folder information. The other copies only have the location and the hash. The representative is picked with a preference for `.claude/skills/` (the folder where Claude Code looks for skills in a repo), so the location of the different skills is biased. That is why the location mix in section 3 uses all occurrences. Folder information and commit history only describe the representative's repo.

How the tables connect.

- artifacts and repos: `repo_full_name` in artifacts matches `full_name` in repos.
- artifacts and artifact_siblings: `repo_full_name` and `path` in artifacts match `repo_full_name` and `artifact_path` in artifact_siblings. Only the representative has siblings.
- Copies of the same skill have the same `file_sha`, and `dedup_primary = 1` marks the representative.
- mining_runs does not connect to the others. It only logs how the data was collected.

What is missing. There is no skill domain or category. I cannot tell which copy is the original or who copied from whom. There is no usage data, so I cannot tell if an agent ever uses a skill. Stars belong to the repo, not the skill. There is no label for human or AI

authorship, and nothing about what the bundled scripts do. The dataset also has no bug reports, issues or comments. The only text is the SKILL.md and the small bundled text files.

Easy to answer: counts, sizes, locations, bundling and exact copies. **Hard to answer:** causes, quality, real usage and near duplicates.

Things to know. 1,544 representatives have no folder listing. The has_scripts column says yes or no if a skill's folder has scripts, and the paper does not say how it is decided. It is a subset of my file-extension rule (a skill has scripts if its folder has a file ending in .py, .sh, .bash, .js, .mjs, .cjs, .ts, .rb, .go, .rs, .ps1, .bat, .java, .php, .pl or .lua): 11,507 skills (5.1% of the skills with script files by extension) have script files but has_scripts is 0.

3. Size metrics

The 3.8M files are 1.88M different skills. The bundled files add 5.9M more files. They are 200.8 GB on GitHub, but the dataset itself is only about 13 GB because it stores the text of just 59.7% of them.

Quantity	Value
File occurrences / different skills	3,797,117 / 1,877,981 (50.5% of the files are copies)
Repos / owners	282,200 / 195,841
Location, all files	canonical (.claude/skills/) 373,652 (9.8%); skills-dir 2,102,053 (55.4%); other 1,321,412 (34.8%)
Valid front matter, different skills	1,625,701 (86.6%)
Skills with scripts (has_scripts), out of 1,876,437 with a folder listing	214,507 (11.4%). Nested skills 198,505 of 1,851,519 (10.7%). Root-level SKILL.md 16,002 of 24,918 (64.2%)
Skills with reference files	485,979 (25.9%). Nested 25.0%, root-level 96.3%
Bundled files / folders	5,858,945 files (200.8 GB on GitHub) / 1,405,920 folders. 3,497,752 files (59.7%) have stored text

Quantity	Value
Skills with commit history / author codes	458,548 / 95,881 (anonymised, includes bots)
Text per skill (text sample, 47,160 skills)	about 970 tokens and 186 lines per skill

Root-level skills (SKILL.md at the top of the repo) are 1.3% of the skills. Their folder is probably the whole repo, so their bundling numbers are inflated. I use nested skills for the comparisons in section 6.

All four variables have a long tail, most values are small and a few are very large. So I show the median next to the mean.

Statistic	Skill body (characters)	Bundled file (bytes)	Commits per skill file	Skills per first-commit author
n	1,877,981	5,858,945	458,548	92,395
mean	6,958	34,273	2.23	4.96
median	4,600	3,558	1	2
std	9,160	626,178	5.77	55.67
variance	8.39e7	3.92e11	33.3	3,099
min	0	0	1	1
q25	2,129	1,221	1	1
q75	8,664	8,734	2	4
max	355,781	104,857,563	708	12,254
skew	7.55	75.3	27.7	148
kurtosis	132	8,190	1,317	28,663

Commits and authors. Commit history exists for 458,548 skills. At least half of them have a single commit (median 1, mean 2.23), and a few have hundreds (max 708). 85.3% of the first commits are from users, 1.0% from bots (4,643) and 13.7% have no author type. The 92,395 first-commit authors have a median of 2 skills each, but the 10 most active authors hold 8.6% of the skills with history, and one author alone has 12,254.

4. Distributions

Most skills are copied zero times, but a few repos hold a big share of the files. 79.3% of the different skills appear once, and 312 repos hold 36.7% of all skill files.

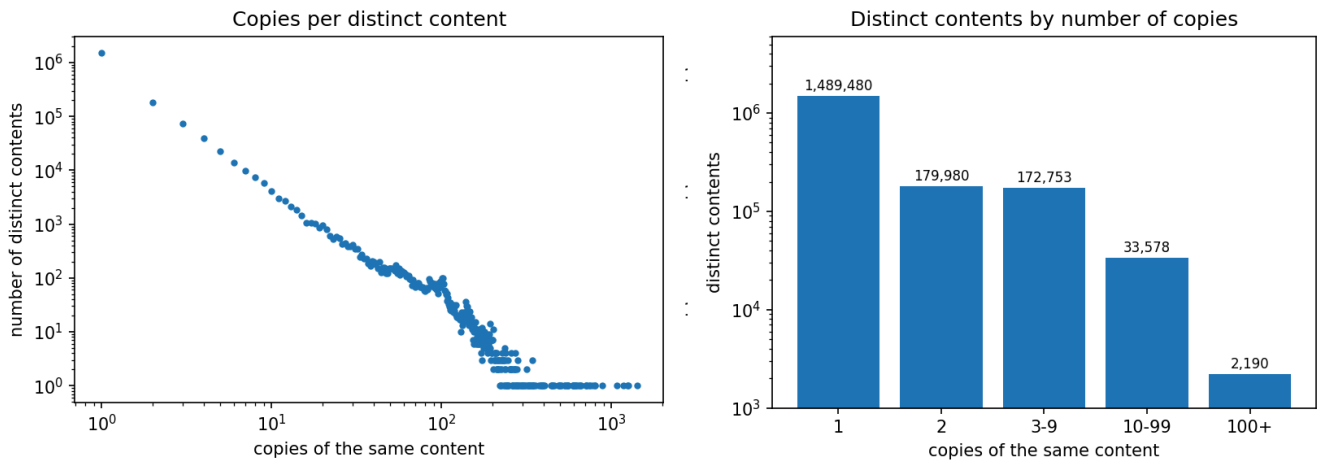


Figure 1. How many copies each skill has. 79.3% of skills appear once and 83.7% are in only one repo. The most copied skill has 1,409 copies in 951 repos. On the log-log plot the points fall close to a straight line over about two decades, so there is a long tail. I did not fit a power law.

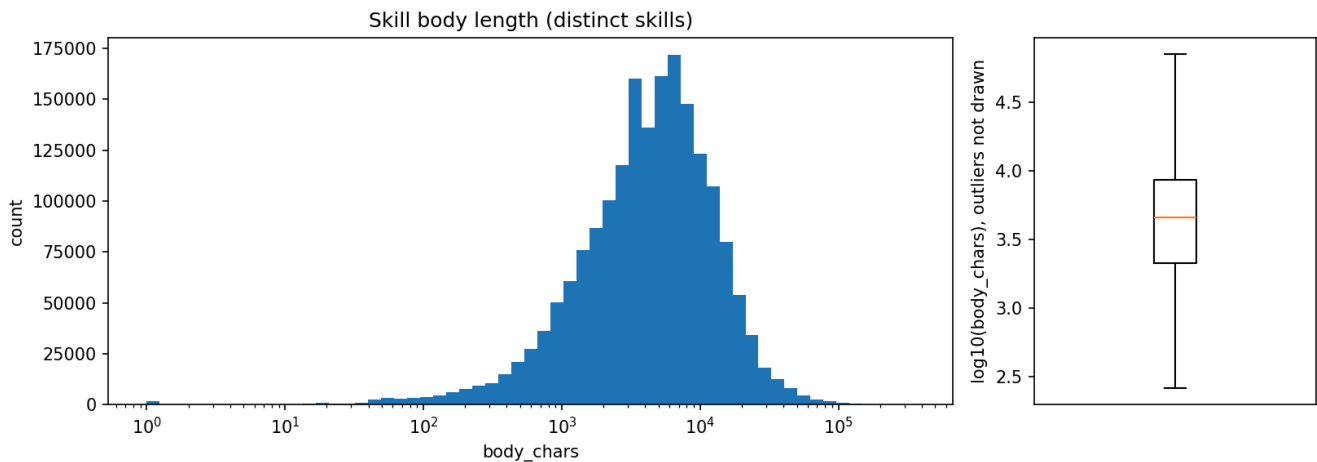


Figure 2. Length of the skill text (log scale). The median is 4,600 characters and most skills are between 1,000 and 10,000. There are a few almost empty skills and a long tail up to 355,781. In the box plot the axis is $\log_{10}$ of the length, so the orange median line at about 3.66 is 4,600 characters.

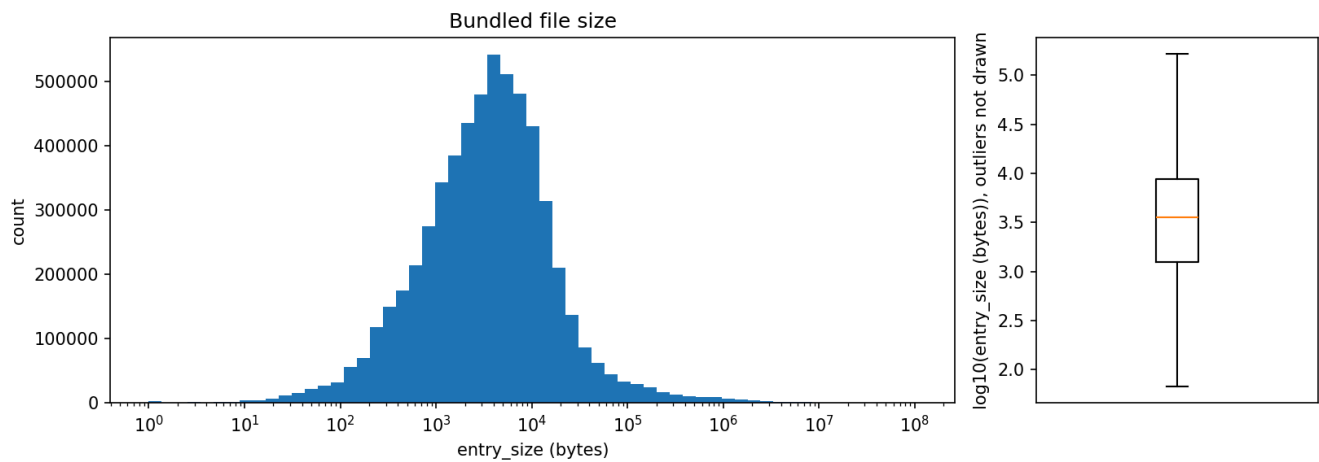


Figure 3. Size of bundled files (log scale, zero-byte files left out). The median is 3.6 KB and the mean is 34 KB. The biggest file is 100 MB, about the size limit. A few very large files make up most of the 200.8 GB. In the sample, the largest were an installer, archives and model weights.

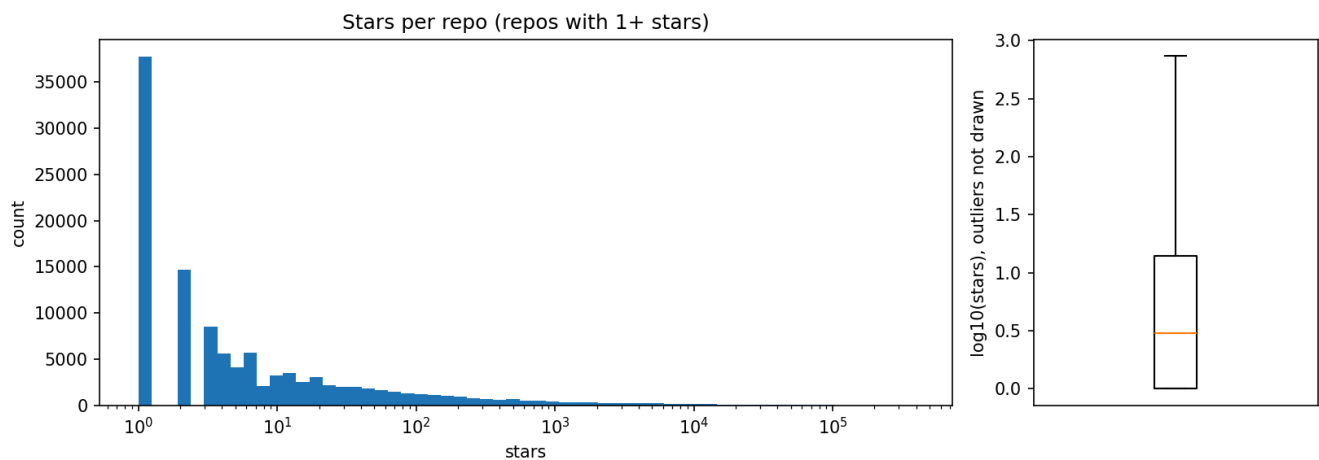


Figure 4. Stars per repo (log scale, repos with at least one star). 59.3% of repos have no stars. The median among repos with stars is 3.

Number of skill files per repo, all repos:

Skill files in repo	Repos	Share of all skill files
1	121,427	3.2%
2 to 5	86,086	6.9%
6 to 20	53,915	14.7%
21 to 100	17,396	18.2%

Skill files in repo	Repos	Share of all skill files
101 to 1,000	3,064	20.4%
1,001 or more	312	36.7%

5. Traceability

Most bundled files are Markdown, and about one in four skills links to a real website.

Bundled files. There are 5,858,945 bundled files, and 3,497,752 of them (59.7%) have stored text. By extension, .md is the biggest with 2,804,074 (47.9%), then .py 619,273, .json 371,240, .ts 234,527, .sh 141,203, .js 138,471, .yaml 137,330 and .png 102,304. There are 1,380,580 script files in about 226,000 skills, and 85.8% of those skills use only one script language.

Script language	Files	Share
Python	619,273	44.9%
TypeScript	234,527	17.0%
JavaScript	208,517	15.1%
Shell	141,474	10.2%
Go	97,097	7.0%
Rust	27,181	2.0%
PowerShell	14,657	1.1%
Java	13,971	1.0%
PHP	13,230	1.0%

URLs (text sample, 47,160 skills). 24.6% of the skills have a link to a real website (11,591 skills), and there are 70,954 links in total. I removed placeholders like example.com and localhost with a simple filter. The most common sites, by number of skills that mention them, are github.com (4,702), raw.githubusercontent.com (412), w3id.org (329), edamontology.org (324), learn.microsoft.com (222), arxiv.org (180), rube.app (174) and

npmjs.com (148). The w3id.org and edamontology.org counts are almost the same, and 324 of the 329 skills that mention w3id.org also mention edamontology.org, so they probably come from one copied template.

Do skills name their own files? (text sample, nested skills with at least one bundled file, 16,489 skills). 64.3% of them name at least one of their bundled files in the SKILL.md text, and 39.7% of all their bundled files are named. For skills with at least one script file (5,112 skills), 70.5% name at least one script, and 36.6% of the script files are named. So about a third of the skills with bundled files never name any of them. This is a simple text match on the file name, so it misses mentions by folder name and can count common names like README.md by accident.

GitHub links (text sample, 47,160 skills). 10.9% of the skills link to github.com (5,135, counting any text with github.com/, so a bit higher than the 4,702 above). 1.9% link to a file or folder in a repo (900), 0.9% to a raw file (414) and 0.3% to an issue or pull request (159).

Language (first 20,000 skills of the text sample, automatic detection). 90.0% are English, 1.6% Chinese, 1.4% German, 1.2% Japanese, 1.0% Portuguese, 0.9% Norwegian, 0.8% Spanish, 0.7% Vietnamese and 0.6% Korean. The detector is not good on short or mixed text, and Norwegian and Catalan look wrong. I have not checked any by hand yet.

Vocabulary (same sample). There are 1,169,098 different words in 45.9M words and 8.8M lines, about 970 words per skill. A word here means a lowercase run of letters, digits and underscores. The vocabulary gets bigger when the sample gets bigger.

Over time. Only 83 of 451,038 first commits are before October 2025, when the format started. New skills per month go from 1,214 in October 2025 to about 85,000 in both May and June 2026. July 2026 is only part of a month.

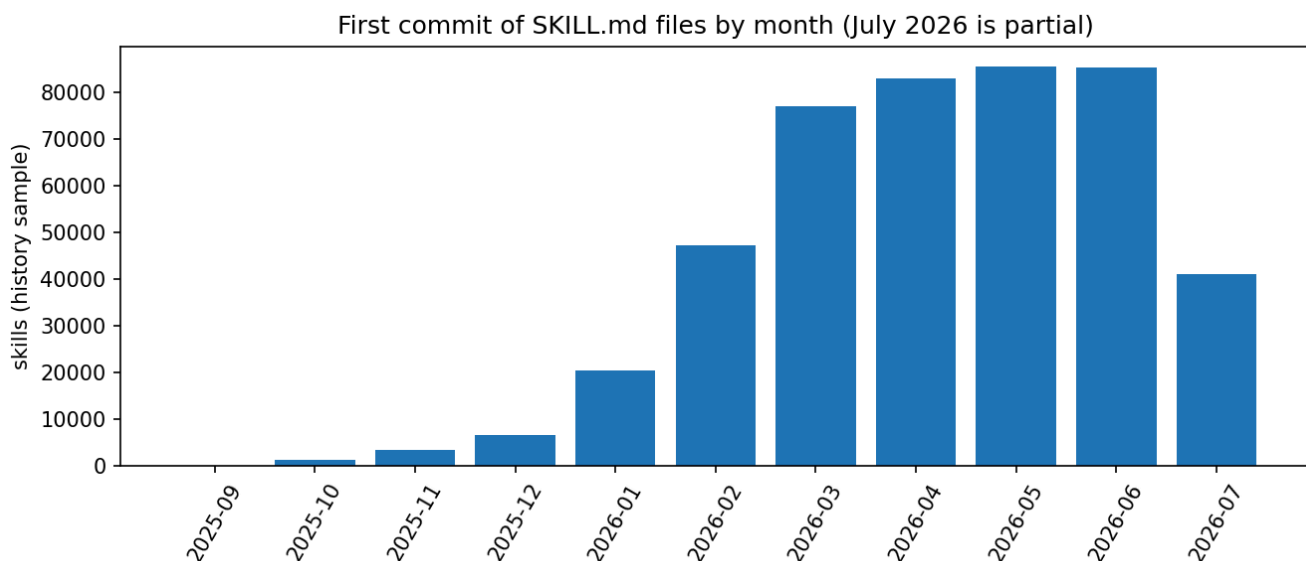


Figure 5. Month of the first commit of the SKILL.md file, for the representative copy of each skill that has history, exact SKILL.md name only (n = 451,038). This is not a random sample.

6. Focused analysis: do skills that bundle scripts spread differently?

Skills with scripts are only slightly more likely to be copied into many repos, and the answer changes with how I define a collection repo.

Definitions. A skill has scripts when the has_scripts flag is 1 (it comes from the representative's folder). Spread is the number of different repos that hold the same skill. A collection repo has 100 or more skill files, widely spread means 10 or more repos, and nested means not a root-level SKILL.md. I fixed the main comparison and the cutoffs before I ran the full data. An odds ratio (OR) above 1 means skills with scripts are more likely to spread.

Setting	n: no scripts / scripts	In 2+ repos	OR (95% CI)	In 10+ repos	OR (95% CI)
All repos, all skills (naive)	1,661,930 / 214,507	16.36% vs 15.78%	0.96 (0.95 to 0.97)	1.30% vs 1.99%	1.54 (1.49 to 1.59)
All repos, nested	1,653,014 / 198,505	16.44% vs 16.82%	1.03 (1.02 to 1.04)	1.31% vs 2.15%	1.66 (1.60 to 1.71)
MAIN: exclude repos with 100+ files, nested	997,335 / 126,568	9.16% vs 9.89%	1.09 (1.07 to 1.11)	0.70% vs 0.81%	1.15 (1.08 to 1.23)
Exclude repos with 20+ files	613,976 / 79,606	7.31% vs 7.93%	1.09 (1.06 to 1.12)	0.42% vs 0.46%	1.08 (0.97 to 1.20)
Exclude repos with 1000+ files	1,245,414 / 168,409	11.10% vs 12.97%	1.19 (1.18 to 1.21)	1.33% vs 1.86%	1.41 (1.35 to 1.46)
MAIN, exact SKILL.md basename only	984,758 / 124,052	9.17% vs 9.97%	1.10 (1.07 to 1.12)	0.71% vs 0.82%	1.16 (1.09 to 1.24)

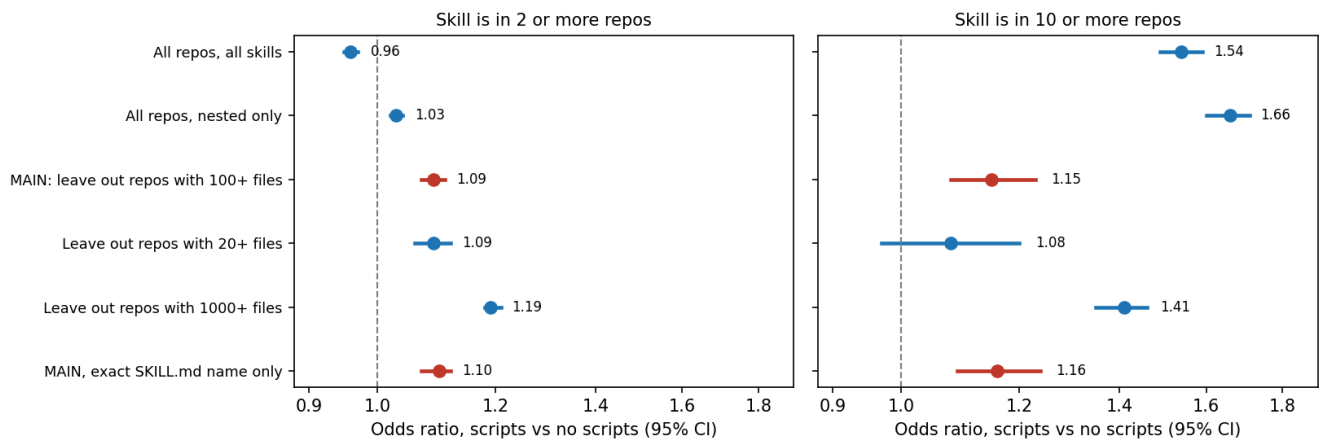


Figure 6. Odds ratios with 95% intervals for each setting in the table (log scale). The dashed line at 1 means no difference. The main comparison is in red.

I do not show p-values. With millions of skills they are always tiny, and skills sit together inside repos, so the intervals are too narrow. The size and direction of the odds ratios matter more.

- **Main result (small).** Outside collection repos, skills with scripts are slightly more likely to be in 10 or more repos (0.81% vs 0.70%, OR 1.15). The difference is 0.11 percentage points.
- **Not stable.** The OR for 10 or more repos is 1.41 when I only leave out the 312 biggest repos, 1.15 at the main cutoff, and 1.08 when I leave out repos with 20 or more files (the interval includes 1). For 2 or more repos the direction stays the same outside collection repos (OR 1.09 to 1.19), but it is close to 1 when collection repos are counted.
- **Collections do most of the copying.** 75.1% of the copies of widely spread skills without scripts, and 73.5% of those with scripts, are in repos with 100 or more skill files. So collection repos, not scripts, are the main way skills spread.
- **Sample vs full data.** My first run on the 13,000 skill sample suggested an odds ratio near 2 for 10 or more repos. The full data does not show that at the main setting, so I should not use the sample number.

Which skills are widely spread and come with scripts? The most spread are webapp-testing (781 repos), several versions of pdf, pptx, xlsx and docx (about 370 to 660 repos each), systematic-debugging (656), mcp-builder (555), skill-creator (532 for one version), slack-gif-creator (469) and algorithmic-art (458). Many look like Anthropic's public example skills, and a few look like popular community packs. I have not checked where they come from.

Where do they live? Two mirror or registry repos hold 1,952 and 1,251 of them (agent-skills-mirror and claude-skill-registry-data). 13 of the top 20 repos have only 664 to 952 skill files each, but hold 573 to 674 of these skills. That is the size of mukul975/Anthropic-Cybersecurity-Skills (790 files, 674 of them), so one security skills collection was copied. I checked this by comparing the file contents: 17 other repos hold at least half of the collection's 790 skills, and 16 of them hold 72% to 99% (the biggest, costrict-plugins-repo, holds 783 of 790). 13 of the 17 repos were created after mukul975/Anthropic-Cybersecurity-Skills and 4 before it, so that repo may not be the original source, and a repo's creation date does not show when the skills were added. 14 of the 17 have 2 stars or fewer.

7. Limitations and next steps

My findings are associations from exact copy matching, and some of them are exploratory.

- I only count exact copies, so lightly edited copies are missed. Folder information (has_scripts, siblings) comes from one representative copy of each skill, and that copy is picked with a preference for .claude/skills/.
- The paper does not say how has_scripts is decided. My comparison is an association, and the topic of a skill and the collection repos could both change the result.
- Words like skill, copy, script and spread have a specific meaning here (section 1). A different definition could change the numbers.
- I looked at several cuts of the data. Only the main comparison was planned before the run, and the others are exploratory.
- The language detection and the URL filter are simple, and I have not checked them by hand.
- The commit history is only for a subset, and it is not random. The dataset only has what GitHub code search finds (default branches, files under 384 KB, recently active repos).

Next steps. I want to find the original source of the security skills collection, because 4 of the 17 repos that copy it were created before the one I used as the source. I also want to match the widely spread skills against the public example skills repo using file_sha, to see where they come from. I want to read what the widely spread scripts do, for example network calls and shell commands. I also want to define skill topics, so I can study one specific area instead of the whole dataset. And should I ask the dataset authors how has_scripts is decided?

Sources

- [GitSkills: A Dataset of Agent Skills on GitHub \(arXiv:2608.10906\)](#)
- [GitSkills on Hugging Face \(Parquet mirror\)](#)
- [MSR 2027 Mining Challenge call](#)
- [GitSkills sample repository](#)